

Liancheng Gong (Krystal)

College Park, MD 20742

gonglc@umd.edu
krystalgong.github.io

EDUCATION

- 2025–
University of Maryland, College Park, MD
Doctor of Philosophy in Information Studies, GPA: 4.00/4.00
Advised by Assistant Professor Julia Mendelsohn
- 2022–2024
University of Pennsylvania, Philadelphia, PA
Master of Science in Engineering in Data Science, GPA: 3.90/4.00
- **Math:** *Big Data Analytics, Statistics for Data Science*
 - **Computer Science:** *Database and Information Systems, Intro to Software Development*
 - **Artificial Intelligence:** *Principles of Deep Learning, Artificial Intelligence, Ethical Algorithm Design*
- 2018–2022
New York University, New York City, NY | Abu Dhabi, UAE | Shanghai, China
Bachelor of Science in Data Science, GPA: 3.81/4.00
- **Math:** *Probability and Statistics, Ordinary Differential Equations, Introduction to Math Modeling*
 - **Core Data Science:** *Data Structures, Machine Learning, Information Visualization, NLP, Algorithms*
 - **Human-Centered:** *Decision Making Under Uncertainty, Human-Centered Data Science*

HONORS AND AWARDS

- 2025–2026
Dean’s Fellowship, University of Maryland
- 2022
Magna Cum Laude, New York University
- 2019–2022
Dean’s List for Academic Year, New York University

PUBLICATIONS

- 2026
Abhinav Rao, **Liancheng Gong**, Bin Hu, Atharva Naik. *An Emergent Mirage: Is Emergent Misalignment and Realignment Indeed a Robust Phenomenon?* Preprint *arXiv:2607.09053*. [Link] [under review]
- 2026
Liancheng Gong, Zhiyang Wang, Yiwei Xu, Julia Mendelsohn. *Persuasion Index: A Theory-Guided Framework for Persuasion Analysis*. Preprint *arXiv:2606.14580*. [Link] [under review]
- 2026
Liancheng Gong, Wang Bill Zhu, Jesse Thomason, Li Zhang. *Iterative Formalization and Planning in Partially Observable Environments*. In Findings of the Association for Computational Linguistics (Findings of ACL).
- 2026
Liancheng Gong, Daniel J. Hopkins, Samuel Wolken. *The Limits of De-politicizing—and Also of Annotation: A Case Study in Russian Newspaper Posts on Social Media 2016–2024*. Proceedings of the International AAAI Conference on Web and Social Media (ICWSM).

CONFERENCE PRESENTATIONS

- 2026
“Persuasion Index: A Theory-Guided Framework for Persuasion Analysis.” **Liancheng Gong**, Zhiyang Wang, Yiwei Xu, Julia Mendelsohn.
- Mid-Atlantic Student Colloquium on AI, Language, and Learning (MASC-ALL). Washington, D.C. April 25.
 - International Conference on Computational Social Science (IC2S2). Burlington, VT. July 28–31.
- 2026
“GeopoliScope: Tracing Geopolitical Stance Formation in Language Model Hidden Representations.” Ashwath David, **Liancheng Gong**, Julia Mendelsohn.
- Mid-Atlantic Student Colloquium on AI, Language, and Learning (MASC-ALL). Washington, D.C. April 25.
 - International Conference on Computational Social Science (IC2S2). Burlington, VT. July 28–31.
- 2026
“Iterative Formalization and Planning in Partially Observable Environments.” **Liancheng Gong**, Wang Bill Zhu, Jesse Thomason, Li Zhang. Association for Computational Linguistics (ACL). San Diego, CA. July 2–7.
- 2026
“The Limits of De-politicizing—and Also of Annotation: A Case Study in Russian Newspaper Posts on Social Media 2016–2024.” **Liancheng Gong**, Daniel J. Hopkins, Samuel Wolken. International AAAI Conference on Web and Social Media (ICWSM). Los Angeles, CA. May 27–29.

REVIEWING

2026 IC2S2, ACM TIST, AAAI

2025 AAAI

FUNDING

2026 CRA-WP Grad Cohort Workshop

Funding source: CRA Scholarship

RESEARCH EXPERIENCE

2024–2025 Drexel University, Department of Computer Science, Philadelphia, PA

Researcher, working with Assistant Professor Harry Zhang

- Structured **Large Language Model (LLM)** outputs into **Planning Domain Definition Language (PDDL)** domain and problem files to generate verifiable plans.
- Iteratively refined outputs using environmental feedback to improve generation accuracy.
- Categorized recurring errors across trials, experimenting with solutions including memory integration and partial file editing to enhance model robustness.

2024–2025 University of Pennsylvania, Department of Political Science, Philadelphia, PA

Research Coordinator (full-time), working with Professor Daniel Hopkins

- Analyzed polls and exit polls for FiveThirtyEight's 2024 presidential election coverage.
- Investigated Democratic and Republican advertisement campaigns across Meta, Instagram, and Google; examined topic trends in Russian media outlets, such as anti-government narratives.
- Scraped and processed 2M+ VKontakte posts from 19 media outlets via **HPC**, created feature vectors using **TF-IDF** and **BERT** embeddings, and developed a **logistic regression model** to identify political posts, increasing the sample's political content from 10% to 50%.
- Annotated Russian blurbs via **OpenAI API**, including prompt engineering for annotation and translation, achieving 90% alignment between GPT and human annotations on political features.
- Classified symbolic rhetoric by generating embeddings and fine-tuning on **BERTweet** to detect topics like "us vs. them" and abstract themes.

2024 University of Pennsylvania, Computational Social Science Lab, Philadelphia, PA

Data Scientist, working with Yuxuan Zhang and Regina (Jeanne) Ruane

- Designed a master list to annotate and analyze bias in media discussions on topics like Biden's age and NATO, delivering findings to the **dashboard**.
- Automated **extract, transform, and load (ETL)** processes for streaming Nielsen data, improving data consistency and minimizing manual handling.
- Managed an **AWS-hosted database**, ensuring data accuracy and schema consistency; standardized network names across large datasets for enhanced data cohesion and analysis.

2023 University of Pennsylvania, Computational Social Science Lab, Philadelphia, PA

Graduate Research Assistant

Atlas, working with Mark Whiting

- Segmented and mined information from PDFs, leveraging **OpenAI's question-answering API** and context-window techniques, e.g., Dynamic Memory Networks and Parallel Context Windows.
- Pioneered an **LLM** approach by refining prompts and GPT models, achieving superior comprehension and context alignment.
- Sourced a robust dataset of 300+ papers and crafted 300+ precise questions, covering all critical analysis units and elevating the accuracy and depth of the literature analysis.

Human Mobility Data for Epidemiology, working with Jorge Barreras Cortes

- Developed **Susceptible-Exposed-Infected-Recovered models** on contact networks utilizing **Poisson Distribution** and **Tensor** methods to predict virus spread, guiding intervention strategies.
- Managed **ETL** of 100GB+ databases via **AWS RDS and EMR**; conducted cross-analysis on POI distribution and mobility indicators, ensuring model applicability and accuracy.
- Optimized the **DBSCAN** clustering algorithm on population data, providing insights into area-specific visit patterns.

PROJECTS

Machine Q&A on Chinese Judicial Reading Comprehension [Code] [PDF]

Developed a Span-Extraction Machine Reading Comprehension (MRC) system.

Colorize Images with Deep Learning [Code] [PDF]

Compared ResNet, GAN, and VAE approaches to colorize greyscale images.

Customer Lifecycle, NYU X HSBC [Code] [PDF]

Segmented customers, predicted inactivity, and provided targeted recommendations for HSBC.

A Model to Predict the Maximum Car Speed in the Rain [Code] [PDF]

Quantified driver visibility and calculated the maximum safe speed under rainy conditions.

TEACHING EXPERIENCE

2022–2023

Fife Academy Coding Club, University of Pennsylvania, Philadelphia, PA

Coding Instructor

- Developed and led a modular coding curriculum for 100+ K–12 students, with a 100% positive feedback rate, delivering weekly lectures, hands-on labs, and individual feedback.

INDUSTRY EXPERIENCE

2021

Tencent Technology (Shenzhen) Co., Ltd, Shanghai, China

Data Scientist, Platform and Content Group

- Enhanced user segmentation and ad CTR by 5% through analysis of 100M+ user records (Python, MySQL); developed ARIMA (0.073 RMSE) for activity forecasting and used NLP on search logs, increasing retention by 4% via A/B testing.

2020–2021

Shanghai Urban Real Estate Appraisal Co., Ltd, Shanghai, China

Data Analyst, Data R&D Department

- Developed predictive models (Regression, Random Forest, XGBoost) with 80% R^2 for rental pricing, supported by literature review, a survey with 300+ responses, and a 2,000+ entry database from field visits and web scraping.

COMPUTER SKILLS

Programming Languages: Python, R, SQL, Java

Frameworks and Libraries: NumPy, Pandas, SciPy, Scikit-Learn, PySpark, Seaborn, Matplotlib, PyTorch, TensorFlow

Tools and Platforms: AWS (RDS/SageMaker/S3/EMR), MySQL, MongoDB, Tableau, Power BI, Git, Linux, Stata, Matlab, ETL